Assignment 1

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Abstract

Abstract: Non-negative Matrix Factorisation (NMF) is a technique for image denoising and feature extraction in computer vision applications, yet standard NMF formulations with Frobenius norm-based loss functions are sensitive to outliers and corrupted input data. This study presents an evaluation of three robust NMF variants designed to handle noisy image data: L1-Regularised Robust NMF, L2,1-Norm Based NMF, and Hypersurface Cost Based NMF (HCNMF). We evaluate these algorithms on two greyscale face image datasets—the ORL Dataset of Faces (400 images across 40 subjects) and the Extended YaleB Dataset (2,414 images of 38 subjects under varying illumination conditions)—under two noise models: salt-and-pepper noise at densities of 0.2 and 0.4, and block occlusion noise with block sizes of 10×10 and 12×12 pixels. Performance is assessed using three metrics: Relative Reconstruction Error (RRE), clustering accuracy (ACC), and Normalised Mutual Information (NMI), with each experimental configuration executed five times using random 90% data subsets. Our results demonstrate that L1-Regularised Robust NMF achieves the lowest median reconstruction error and highest clustering accuracy and NMI scores, particularly on salt-and-pepper noise through its sparse error modelling. L2,1-NMF proves most effective for handling block occlusions through its sample-level error weighting and demonstrates the lowest standard deviation in clustering accuracy across experimental runs. HCNMF, whilst exhibiting higher reconstruction error by Frobenius norm standards, shows competitive clustering performance and performs best on the Extended YaleB dataset with extreme illumination variations. These findings provide guidance for selecting robust NMF methods based on noise characteristics, dataset properties, and application requirements.

1 Introduction

It is well known within the data science community that clean real-world data is incredibility hard to find. Within computer vision applications most image data is plagued with noise due to varying illumination, shadows and occlusion. In particular the medical industry has had decades of advancement in imaging technology, however, many of the medical images can be affected by distortions like white noise inherent to the medical device and also the type of noise can vary between devices [6]. Being able to effectively denoise image data and improve machine learning performance on noisy dataset will be essential in advancing computer vision applications such as diagnosis through medical images.

Non-negative matrix factorisation (NMF) is a popular method to deal with noisy image data and is know for its excellent performance in real applications and has simple theoretical interpretations [8]. Unlike traditional matrix factorisation where a matrix is decompressed into smaller matrices, NMF has a special condition where all the elements of the matrices must be non-negative. This is the particularly reason why it's useful for image data as pixels are either zero or positive.

The fundamental principle behind NMF is the decomposition of a data matrix V into smaller matrices, usually two, W and H where the elements of the matrices are non-negative. This decomposition acts as a denoising mechanism which effectively removes high-frequency noise components from the main signal patterns within the image. The standard NMF is known to be not robust to outliers and this the reason why we test robustness of three other NMF variants on two image datasets.

In this paper the three NMF algorithms evaluated are L1 – Norm Regularized Robust NMF, Hypersurface Cost Based NMF (HCNMF) and L2,1 – Norm Based NMF. These algorithms will be evaluated on two grey scale real-word datasets. The ORL "Dataset of Faces" that contains 400 images of 40 distinct subjects and the Extended YaleB dataset that contains 2414 images of 38 subjects. All the images will be manually adjusted to be the same size and aligned.

2 Previous work

Non-negative Matrix Factorisation (NMF) has been extensively studied since its introduction by Lee and Seung [5], with numerous variants developed to address limitations of the standard approach. This section reviews the main categories of NMF methods relevant to our robustness analysis, focusing on their advantages and disadvantages in handling corrupted data.

2.1 Standard NMF and its limitations

Classical NMF factorises a non-negative data matrix $\mathbf{X}$ into non-negative factors $\mathbf{W}$ and $\mathbf{H}$ by minimising either squared-error (Frobenius norm) or Kullback-Leibler (KL) divergence using multiplicative update rules [5]. While effective under small, approximately Gaussian noise conditions, the quadratic loss function in standard NMF is highly sensitive to outliers. A small fraction of large-magnitude corruptions, such as salt-and-pepper noise or block occlusions, can dominate the objective function, leading to distorted factorisation results and degraded reconstruction quality.

Advantages: Computationally efficient, well-established convergence properties, interpretable parts-based representation.

Disadvantages: Sensitive to outliers and gross corruptions, poor performance under impulsive noise, limited robustness to structured occlusions.

2.2 Locality and sparsity-enhanced NMF

Local Non-negative Matrix Factorisation (LNMF) and its variants introduced spatial locality constraints to encourage more localised, parts-based representations [1]. Sparse NMF methods incorporate L_1 regularisation on the factor matrices to promote sparsity in the learnt representations. While these approaches improved spatial locality and sometimes enhanced classification accuracy on face recognition tasks, they do not directly address the fundamental issue of robustness to impulsive or structured corruptions.

Advantages: Improved spatial locality, enhanced interpretability, better feature selection through sparsity.

Disadvantages: Do not specifically target noise robustness, additional hyperparameter tuning required, computational overhead from regularisation terms.

2.3 Robust loss functions for NMF

2.3.1 $L_{2,1}$-norm NMF

$L_{2,1}$-norm NMF replaces the traditional Frobenius loss with a mixed $L_{2,1}$ norm that measures Euclidean reconstruction error per sample and sums these errors linearly across the dataset [4]. This approach addresses sample-level outliers by ensuring that large per-sample errors grow only linearly rather than quadratically, preventing a few badly corrupted images from dominating the optimisation objective.

Advantages: Effective against sample-level outliers, maintains computational tractability through iteratively re-weighted least squares, theoretically well-motivated.

Disadvantages: Less effective for pixel-level sparse corruptions, requires careful weight updating strategies, may converge slowly in practice.

2.3.2 Hypersurface-cost NMF (HCNMF)

HCNMF employs a bounded-influence, differentiable loss function that exhibits quadratic behaviour for small residuals but transitions to linear growth for large residuals [3]. This M-estimator-like approach caps the influence of very large errors without explicitly modelling a sparse error matrix, providing robust estimation through gradient-based optimisation with line search.

Advantages: Robust estimation without explicit sparse error modelling, differentiable objective function enabling gradient-based optimisation, computationally efficient.

Disadvantages: More complex optimisation procedure than standard NMF, potential local minima issues, less interpretable than explicit error separation models.

2.4 Sparse error modelling approaches

2.4.1 L_1-regularised robust NMF

L_1-regularised Robust NMF explicitly models corruption as a sparse error matrix $\mathbf{E}$ by decomposing $\mathbf{X} \approx \mathbf{WH} + \mathbf{E}$ and penalising $\mathbf{E}$ with an L_1 norm [9]. This formulation effectively separates impulsive pixel noise and spatially sparse occlusions into the error matrix $\mathbf{E}$ while capturing clean low-rank structure in the product $\mathbf{WH}$.

Advantages: Excellent performance on salt-and-pepper noise, effective for spatially sparse but contiguous corruptions (block occlusions), clear separation of clean structure and noise.

Disadvantages: Increased computational complexity due to additional matrix variable, requires careful balancing of regularisation parameters, may overfit to noise patterns in some cases.

2.5 Huber loss for robust NMF

The Huber loss is a classic M-estimator that behaves quadratically for small residuals and linearly for large ones, interpolating between ℓ_2 and ℓ_1 penalties. For a residual r and threshold $\delta > 0$,

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\big(|r| - \frac{1}{2}\delta\big), & |r| > \delta. \end{cases}$$

Embedding ρ_δ into NMF yields a robust objective

$$\min_{\mathbf{W},\mathbf{H}\ge 0} \sum_{i,j} \rho_\delta\big((\mathbf{X} - \mathbf{WH})_{ij}\big) \quad (+\text{ regularisers}),$$

which down-weights gross errors while keeping efficient ℓ_2-like fitting on inliers. Compared with pure ℓ_2 (sensitive to outliers) and pure ℓ_1 (can underfit small residuals), Huber provides *bounded influence* with a tunable transition via δ.

Wang et al. [10] integrate the Huber loss into a *sparse graph-regularised* NMF for cancer genomics, coupling robustness with manifold smoothness and sparsity. Their model augments the Huber-reconstruction term with (i) a graph Laplacian penalty to preserve local structure among samples and genes, and (ii) sparsity regularisation to encourage interpretability. Optimisation proceeds via alternating updates with re-weighting and half-quadratic strategies, effectively performing iteratively re-weighted least squares (IRLS) where large residuals receive smaller weights. On cancer datasets, this approach improved subtype separation and biomarker interpretability compared to vanilla NMF and several robust baselines.

Advantages: Balanced robustness through smooth transition between quadratic and linear penalties, tunable sensitivity via δ parameter, well-suited for datasets with mixed noise characteristics.

Disadvantages: Requires careful selection of threshold parameter δ, more complex optimisation than standard NMF, computational overhead from iterative re-weighting.

2.6 Other robust approaches not adopted

Several other robust NMF variants exist but were not selected for this study. Graph-regularised NMF (beyond the Huber-based variant discussed above) incorporates manifold information for enhanced discrimination but introduces additional hyperparameters and increased computational complexity. Semi-supervised and supervised NMF variants leverage label information but require ground truth annotations, which may not always be available. Deep and hierarchical NMF methods capture multi-level structure but significantly increase training and tuning complexity [2].

Alternative robust loss functions such as Cauchy loss and truncated losses also reduce outlier influence, but the selected methods represent well-established approaches covering complementary robustness principles: sample-level down-weighting ($L_{2,1}$-NMF), explicit sparse-error separation (L_1-Robust NMF), bounded-influence costs (HCNMF), and hybrid quadratic-linear penalties (Huber-NMF).

3 Methods

3.1 L1 Regularized Robust Non-negative Matrix Factorization

This section describes our implementation of L1 regularized robust Non-negative Matrix Factorization (NMF), which combines L1 reconstruction loss with L1 regularization to achieve robustness against outliers and promote sparsity in the learned representations.

3.1.1 Problem Formulation

Given a non-negative input matrix $\mathbf{V} \in \mathbb{R}_+^{m \times n}$, our L1 regularized robust NMF seeks to find non-negative factor matrices $\mathbf{W} \in \mathbb{R}_+^{m \times k}$ and $\mathbf{H} \in \mathbb{R}_+^{k \times n}$ that minimize the following objective function:

$$\mathcal{L}(\mathbf{W}, \mathbf{H}) = \|\mathbf{V} - \mathbf{WH}\|_1 + \lambda_1 \|\mathbf{W}\|_1 + \lambda_2 \|\mathbf{H}\|_1 \tag{1}$$

subject to the constraints:

$$\mathbf{W} \geq 0 \tag{2}$$
$$\mathbf{H} \geq 0 \tag{3}$$

where $\|\cdot\|_1$ denotes the L1 norm (sum of absolute values), k is the number of components (latent factors), and $\lambda_1, \lambda_2 > 0$ are regularization parameters controlling the sparsity of $\mathbf{W}$ and $\mathbf{H}$, respectively.

The L1 reconstruction loss $\|\mathbf{V} - \mathbf{WH}\|_1$ provides robustness to outliers by limiting the influence of large reconstruction errors, while the L1 regularization terms promote sparsity in the factor matrices, encouraging interpretable parts-based representations.

3.1.2 Gradient Descent

We employ a simple gradient descent approach to solve the constrained optimization problem. The gradients of the reconstruction loss with respect to $\mathbf{W}$ and $\mathbf{H}$ are computed using the subgradient of the L1 norm:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = -\text{sign}(\mathbf{V} - \mathbf{WH})\mathbf{H}^T \tag{4}$$
$$\frac{\partial \mathcal{L}}{\partial \mathbf{H}} = -\mathbf{W}^T \text{sign}(\mathbf{V} - \mathbf{WH}) \tag{5}$$

where $\text{sign}(\cdot)$ is the element-wise sign function, which serves as the subgradient of the L1 norm.

The factor matrices are updated using a fixed step size $\alpha = 0.0001$ in an alternating fashion:

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \alpha \frac{\partial \mathcal{L}}{\partial \mathbf{W}} \tag{6}$$

$$\mathbf{H}_{t+1} = \mathbf{H}_t - \alpha \frac{\partial \mathcal{L}}{\partial \mathbf{H}} \tag{7}$$

3.1.3 Soft Thresholding for L1 Regularization

The L1 regularization terms are handled through soft thresholding operators applied after each gradient update:

$$\mathbf{W}_{t+1} = \mathcal{S}_{\alpha\lambda_1}(\mathbf{W}_t - \alpha \frac{\partial \mathcal{L}}{\partial \mathbf{W}}) \tag{8}$$

$$\mathbf{H}_{t+1} = \mathcal{S}_{\alpha\lambda_2}(\mathbf{H}_t - \alpha \frac{\partial \mathcal{L}}{\partial \mathbf{H}}) \tag{9}$$

where the soft thresholding operator $\mathcal{S}_\tau(\cdot)$ is defined element-wise as:

$$\mathcal{S}_\tau(x) = \text{sign}(x) \max(|x| - \tau, 0) \tag{10}$$

This operator effectively implements the proximal operator for the L1 penalty, simultaneously performing gradient descent on the reconstruction loss and enforcing sparsity through the regularization terms. The threshold values are $\alpha\lambda_1$ and $\alpha\lambda_2$ for $\mathbf{W}$ and $\mathbf{H}$ respectively, where $\alpha = 0.0001$ is the fixed step size.

3.1.4 Non-negativity Constraints

After each update, non-negativity constraints are enforced through projection:

$$\mathbf{W}_{t+1} = \max(\mathbf{W}_{t+1}, \epsilon\mathbf{1}) \tag{11}$$

$$\mathbf{H}_{t+1} = \max(\mathbf{H}_{t+1}, \epsilon\mathbf{1}) \tag{12}$$

where $\max(\cdot, \cdot)$ is applied element-wise, $\mathbf{1}$ is a matrix of ones, and $\epsilon = 10^{-10}$ prevents numerical instabilities by maintaining strictly positive values.

3.1.5 Initialization Strategy

To ensure good convergence properties, we initialize the factor matrices using a truncated Singular Value Decomposition (SVD) approach:

1. Compute the SVD of $\mathbf{V}$: $\mathbf{V} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$
2. Initialize $\mathbf{W}$ and $\mathbf{H}$ using the first k components:

$$\mathbf{W}^{(0)} = \max(\mathbf{U}_{:,1:k}\sqrt{\mathbf{\Sigma}_{1:k,1:k}}, \epsilon) \tag{13}$$

$$\mathbf{H}^{(0)} = \max(\sqrt{\mathbf{\Sigma}_{1:k,1:k}}\mathbf{V}^T_{1:k,:}, \epsilon) \tag{14}$$

3. If k exceeds the rank of $\mathbf{V}$, pad with small random values drawn from an exponential distribution scaled by the data magnitude

This initialization provides a good starting point that captures the dominant structure in the data while satisfying non-negativity constraints.

3.1.6 Convergence Criteria

The algorithm monitors two convergence criteria:

1. **Objective function convergence**: Stop when relative change in objective falls below tolerance:

$$\frac{|\mathcal{L}_t - \mathcal{L}_{t-1}|}{\mathcal{L}_{t-1}} < \text{tol} \tag{15}$$

2. **Parameter convergence**: Stop when relative changes in both factor matrices are small:

$$\frac{\|\mathbf{W}_t - \mathbf{W}_{t-1}\|_F}{\|\mathbf{W}_{t-1}\|_F} < \text{tol} \tag{16}$$

$$\frac{\|\mathbf{H}_t - \mathbf{H}_{t-1}\|_F}{\|\mathbf{H}_{t-1}\|_F} < \text{tol} \tag{17}$$

where tol $= 10^{-6}$ is the convergence tolerance. The algorithm terminates when either criterion is satisfied or the maximum number of iterations (1000) is reached.

3.1.7 Transform Operation

For transforming new data $\mathbf{V}_{\text{new}}$ to the learned latent space, we fix $\mathbf{W}$ and optimize only $\mathbf{H}_{\text{new}}$ using the same gradient-based approach with L1 regularization. This allows efficient encoding of new samples using the learned basis functions in $\mathbf{W}$.

3.2 HCNMF

The HCNMF replaces the standard NMF Frobenius norm-based loss function with a differentiable hypersurface cost which is asymptotically linear for large residuals and quadratic when near zero making it highly robust to outliers.

3.2.1 Objective Function

$$\min_{W\geq 0, H\geq 0} \mathcal{L}(\mathbf{W}, \mathbf{H}) = \sum_{i=1}^{m} \sum_{j=1}^{n} \left(\sqrt{1 + (\mathbf{V}_{ij} - [\mathbf{WH}]_{ij})^2} - 1 \right). \tag{18}$$

3.2.2 Cost Function

$$\delta(x) = \sqrt{1 + x^2} - 1. \tag{19}$$

When x is small, the cost function is quadratic (since $\sqrt{1+x^2} - 1 \approx \frac{1}{2}x^2$) and when x is large the cost function is linear (since $\sqrt{1+x^2} - 1 \approx |x|$). This help reduce the effects of outliers [11].

3.2.3 Iterative Updates

$$W_{ik}^{(t+1)} = W_{ik}^{(t)} - \alpha_{ik}^{(t)} \frac{\left(WHH^T\right)_{ik}^{(t)} - \left(SH^T\right)_{ik}^{(t)}}{\sqrt{1 + \|S - WH\|}} \tag{20}$$

$$H_{kj}^{(t+1)} = H_{kj}^{(t)} - \beta_{kj}^{(t)} \frac{\left(W^T WH\right)_{kj}^{(t)} - \left(W^T S\right)_{kj}^{(t)}}{\sqrt{1 + \|S - WH\|}} \tag{21}$$

Alpha and beta are step sizes chosen via the Armijo rule which is a line search method that control the step size [7].

3.3 L2,1-Norm Based NMF

The L2,1-Norm Based Non-negative Matrix Factorization (L2,1-NMF) is a robust variant of the standard NMF. It is designed to handle data with sample-specific outliers or corruptions, such as images where some samples are heavily obscured. Unlike the standard NMF, which uses a Frobenius norm (L_2 norm) loss function that is sensitive to large errors due to squaring, the L2,1-NMF employs a loss function that is less sensitive to such outliers. This is achieved by measuring the reconstruction error on a per-sample basis and summing these errors linearly, which prevents a few heavily corrupted data points from dominating the optimization process.

3.3.1 Objective Function

Given a non-negative data matrix $\mathbf{V} \in \mathbb{R}_{+}^{m\times n}$, L2,1-NMF aims to find two non-negative factor matrices, $\mathbf{W} \in \mathbb{R}_{+}^{m\times k}$ and $\mathbf{H} \in \mathbb{R}_{+}^{k\times n}$, by solving the following optimization problem

$$\min_{\mathbf{W}\geq 0,\mathbf{H}\geq 0} ||\mathbf{V} - \mathbf{WH}||_{2,1}$$

where the $L_{2,1}$-norm of a matrix $\mathbf{E} \in \mathbb{R}^{m\times n}$ is defined as the sum of the Euclidean (L_2) norms of its column vectors

$$||\mathbf{E}||_{2,1} = \sum_{j=1}^{n} ||\mathbf{e}_j||_2 = \sum_{j=1}^{n} \sqrt{\sum_{i=1}^{m} E_{ij}^2}$$

Here, $\mathbf{e}_j$ represents the j-th column of the error matrix $\mathbf{E} = \mathbf{V} - \mathbf{WH}$, corresponding to the reconstruction error for the j-th data sample. By not squaring the norm of each column's error, this objective function ensures that the influence of outliers grows linearly rather than quadratically, leading to improved robustness

3.3.2 Optimization and Iterative Updates

The optimization problem for L2,1-NMF is typically solved using an iteratively re-weighted least squares (IRLS) strategy. The $L_{2,1}$-norm objective can be reformulated as a weighted Frobenius norm minimization problem:

$$\min_{\mathbf{W}\geq 0,\mathbf{H}\geq 0} \mathrm{Tr}((\mathbf{V} - \mathbf{WH})\mathbf{D}(\mathbf{V} - \mathbf{WH})^T)$$

where $\mathbf{D}$ is a diagonal matrix whose entries are defined based on the reconstruction error for each data sample (column). The diagonal elements D_{jj} are calculated as follows

$$D_{jj} = \frac{1}{\|(\mathbf{V} - \mathbf{WH})_j\|_2 + \epsilon}$$

where ϵ is a small constant to ensure numerical stability. This formulation assigns a lower weight to data samples with larger reconstruction errors, effectively suppressing the influence of outliers.

With the problem cast as a weighted Frobenius norm minimization, multiplicative update rules can be derived.The factor matrices $\mathbf{W}$ and $\mathbf{H}$ are updated iteratively and alternately until convergence:

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{W}^T\mathbf{VD}}{\mathbf{W}^T\mathbf{WHD} + \epsilon} \tag{22}$$

$$\mathbf{W} \leftarrow \mathbf{W} \odot \frac{\mathbf{VDH}^T}{\mathbf{WHDH}^T + \epsilon} \tag{23}$$

where $\odot$ denotes element-wise multiplication and the division is also element-wise. The algorithm alternates between updating the weight matrix $\mathbf{D}$ based on the current reconstruction error, and then updating the factor matrices $\mathbf{W}$ and $\mathbf{H}$ using these weights.

4 Experiments and Discussions

This section details the experimental setup used to evaluate the robustness of the implemented Non-negative Matrix Factorization (NMF) algorithms. We present and analyze the results, comparing the performance of each algorithm across different datasets, noise types, and noise intensities.

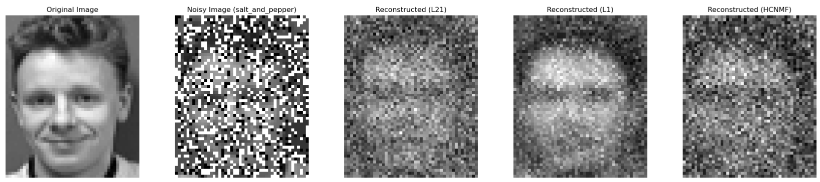

Figure 1: Example of Salt and Pepper noise with the reconstructions from all three algorithms

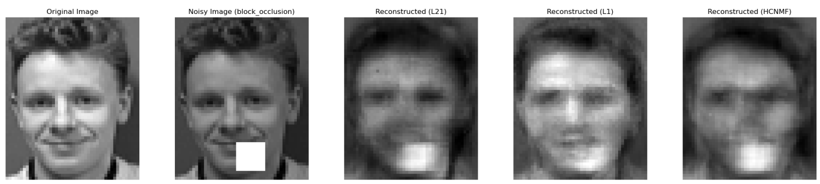

Figure 2: Example of block occlusion noise with the reconstructions from all three algorithms

4.1 Experimental Setup

Our evaluation framework was designed to rigorously test the NMF variants under controlled conditions.

4.1.1 Datasets

The experiments were conducted on two standard grayscale face image datasets:

- **ORL Dataset:** Contains 400 images of 40 distinct subjects. The images feature variations in facial expressions and details against a uniform background.
- **Extended YaleB Dataset:** A more challenging dataset with 2,414 images of 38 subjects, captured under 64 different illumination conditions and 9 poses.

For computational efficiency and to standardize the feature space, all images from both datasets were downsampled by a reduction factor of 2.

4.1.2 Algorithms Compared

We implemented and compared three robust NMF algorithms:

- L1-Regularized Robust NMF (L1-Robust-NMF)
- $L_{2,1}$-Norm Based NMF (L2,1-NMF)
- Hypersurface Cost Based NMF (HCNMF)

4.1.3 Noise Models

To simulate data corruption and test robustness, two types of noise were programmatically added to the images:

- **Salt-and-Pepper Noise:** This model simulates sharp, sparse disturbances by randomly setting a fraction of pixels to maximum (255) or minimum (0) intensity. The experiments used noise densities p of 0.2 and 0.4, with salt-to-pepper ratios r of 0.5 and 0.7.
- **Block Occlusion Noise:** This model simulates real-world occlusions by replacing a rectangular region of the image with a solid white block. Block sizes of 10x10 and 12x12 pixels were used.

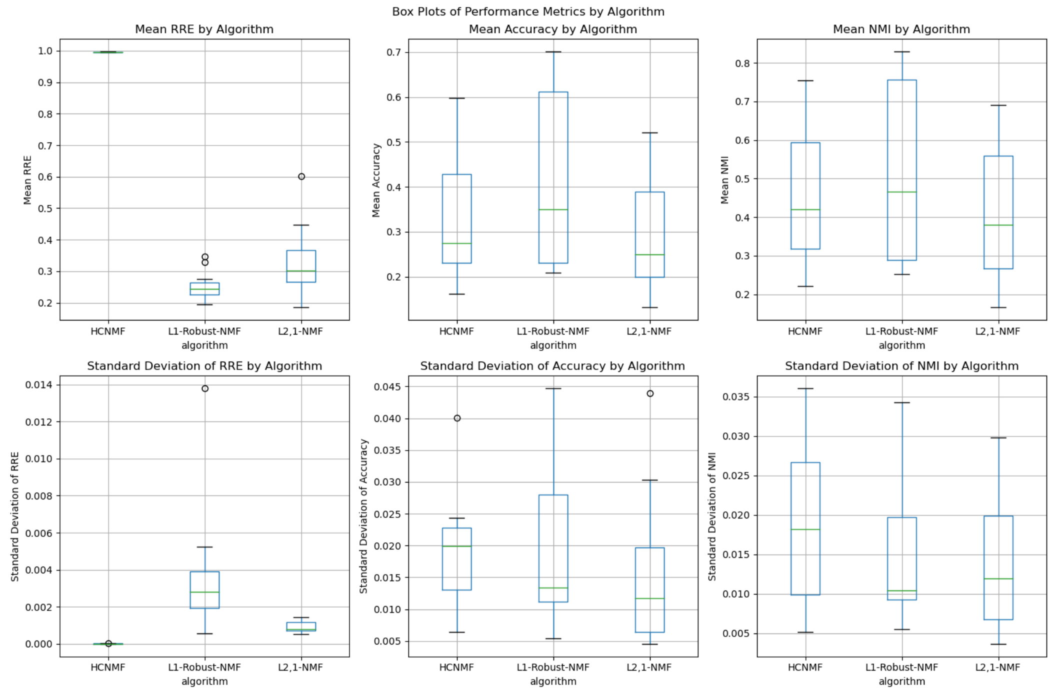

Figure 3: Overall performance metrics (Mean RRE, Accuracy, and NMI) grouped by algorithm.

4.1.4 Evaluation Protocol

The performance of each algorithm was quantified using three standard metrics:

- **Relative Reconstruction Error (RRE):** Measures the fidelity of the reconstruction against the original, clean data, defined as $RRE = \frac{\|\hat{V} - WH\|_F}{\|\hat{V}\|_F}$, where $\hat{V}$ is the original clean data matrix.
- **Clustering Accuracy (ACC):** Evaluates the quality of the learned coefficient matrix H for separating data into its true classes.
- **Normalized Mutual Information (NMI):** Another metric to assess clustering quality based on the mutual information between the predicted and true labels.

To ensure statistical significance, each experimental configuration was executed 5 times. In each run, a random 90% subset of the data was sampled. The results are reported as the mean and standard deviation across these runs. All algorithms were run for a maximum of 200 iterations. The entire experimental suite was parallelized using Python's multiprocessing library for efficiency.

4.2 Overall Performance Analysis

Aggregating the results across all datasets and noise configurations provides a high-level view of each algorithm's general performance and stability.

- **Reconstruction Error (RRE):** The L1-Robust-NMF algorithm demonstrated the lowest median RRE, closely followed by L2,1-NMF. This is consistent with its design, which explicitly models sparse errors, allowing it to effectively separate both salt-and-pepper noise and block occlusions from the underlying image data. In stark contrast, HCNMF yielded a very high RRE, often approaching 1.0. This does not necessarily indicate a bug, as the RRE calculation was shared across all models and HCNMF produced reasonable clustering scores for ACC and NMI. This suggests that while the HCNMF objective function, which

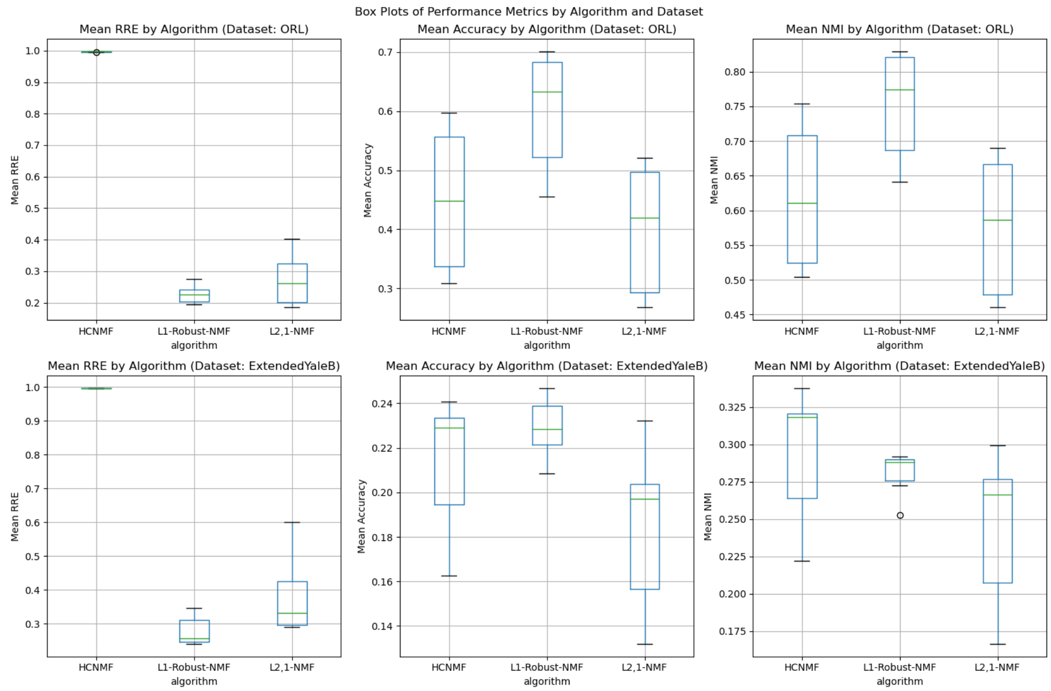

Figure 4: Performance metrics by algorithm, compared across the ORL and ExtendedYaleB datasets.

bounds the influence of large errors, does not prioritize minimizing the Frobenius norm, it still successfully learns a feature space (H) that preserves class separability.

- **Clustering Performance (ACC & NMI):** In terms of clustering, L1-Robust-NMF achieved the highest median Accuracy, while also having the highest median NMI score. This indicates that its superior denoising capability translates directly to learning more discriminative features in the coefficient matrix H. L2,1-NMF was the second-best performer in Accuracy, with HCNMF and L2,1-NMF showing comparable NMI scores.
- **Stability:** The stability of the algorithms, measured by the standard deviation of their scores, varied by metric. L2,1-NMF exhibited the lowest standard deviation in Accuracy, suggesting it is a highly consistent and reliable performer across different data subsets. Conversely, L1-Robust-NMF showed the lowest standard deviation for NMI, indicating its robustness in preserving the mutual information between true and predicted labels. L1-Robust-NMF had the highest standard deviation in RRE, which may reflect its sensitivity to initialization when separating the data and error matrices.

4.3 Performance by Dataset

Significant performance differences were observed between the ORL and Extended YaleB datasets, highlighting the impact of inherent dataset characteristics.

- **ORL Dataset:** All algorithms performed substantially better on the ORL dataset, with L1-Robust-NMF achieving the highest Accuracy and NMI scores. Accuracy values on ORL reached as high as 70%, with NMI scores exceeding 0.8. The controlled environment of the ORL dataset, with less variation in lighting and pose, provides a cleaner baseline for the algorithms to work from.
- **Extended YaleB Dataset:** Performance on this dataset was considerably lower, with maximum accuracy around 25% and NMI just above 0.33. The severe illumination changes in Extended YaleB act as a form of complex, non-sparse noise that challenges all models. Interestingly, on this more difficult dataset, HCNMF emerged as the top performer in both Accuracy and NMI, albeit by a small margin. This suggests that HCNMF's bounded cost

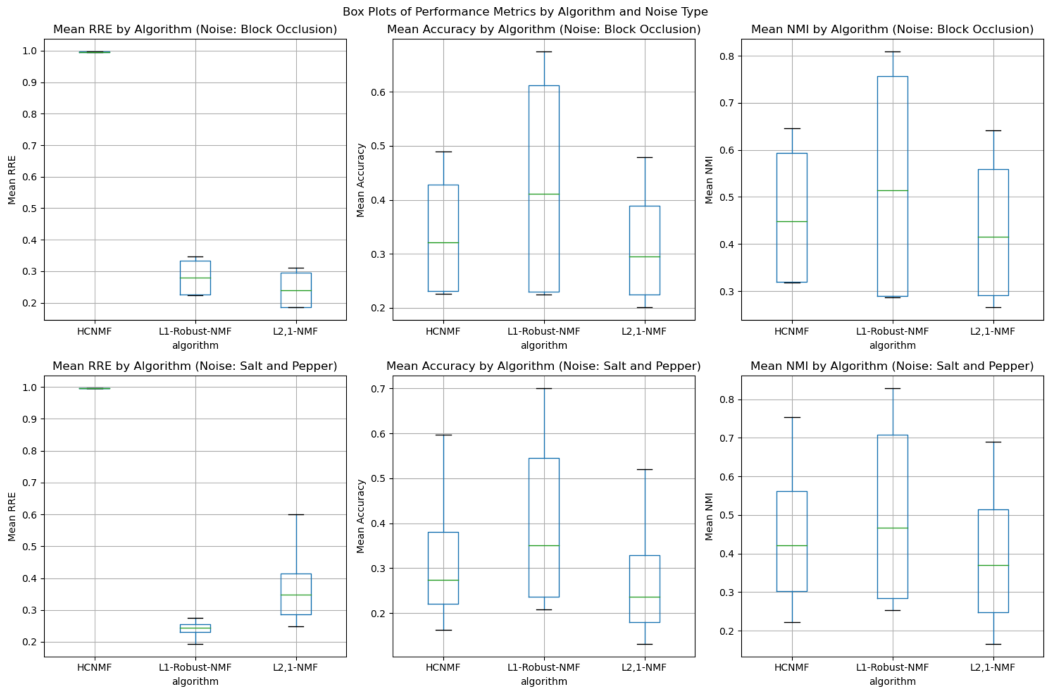

Figure 5: Performance metrics by algorithm, compared across Block Occlusion and Salt and Pepper noise types.

function, which is robust to generic outliers without assuming a specific error structure (like sparsity), may be better suited for handling the combined effects of extreme lighting variations and the manually added noise.

4.4 Performance by Noise Type

The type of noise applied had a notable impact on the relative performance of the algorithms, particularly regarding reconstruction error.

- **Salt-and-Pepper Noise:** Under this condition, L1-Robust-NMF had a lower RRE than L2,1-NMF. This result strongly aligns with theory, as the L1 norm in the objective function is ideally suited to handle sparse, pixel-wise corruptions.
- **Block Occlusion Noise:** Counter-intuitively, with block occlusion noise, L2,1-NMF achieved a slightly lower RRE than L1-Robust-NMF. This suggests that the $L_{2,1}$ norm, which evaluates error on a per-sample (column) basis, is highly effective at down-weighting entire images that are corrupted by large, contiguous occlusions at important locations in the image. The algorithm effectively identifies and reduces the influence of these heavily damaged samples during optimization.

4.5 Impact of Noise Parameters

Across all experiments, the relative performance ranking of the three algorithms remained consistent regardless of the specific noise parameters used. For instance, L1-Robust-NMF consistently outperformed the other algorithms on salt-and-pepper noise whether the noise density p was 0.2 or 0.4. This consistency demonstrates that the observed strengths and weaknesses of each algorithm are fundamental to their design and not artifacts of a specific noise level. This robustness across varying noise intensities reinforces our conclusions about their core mechanisms.

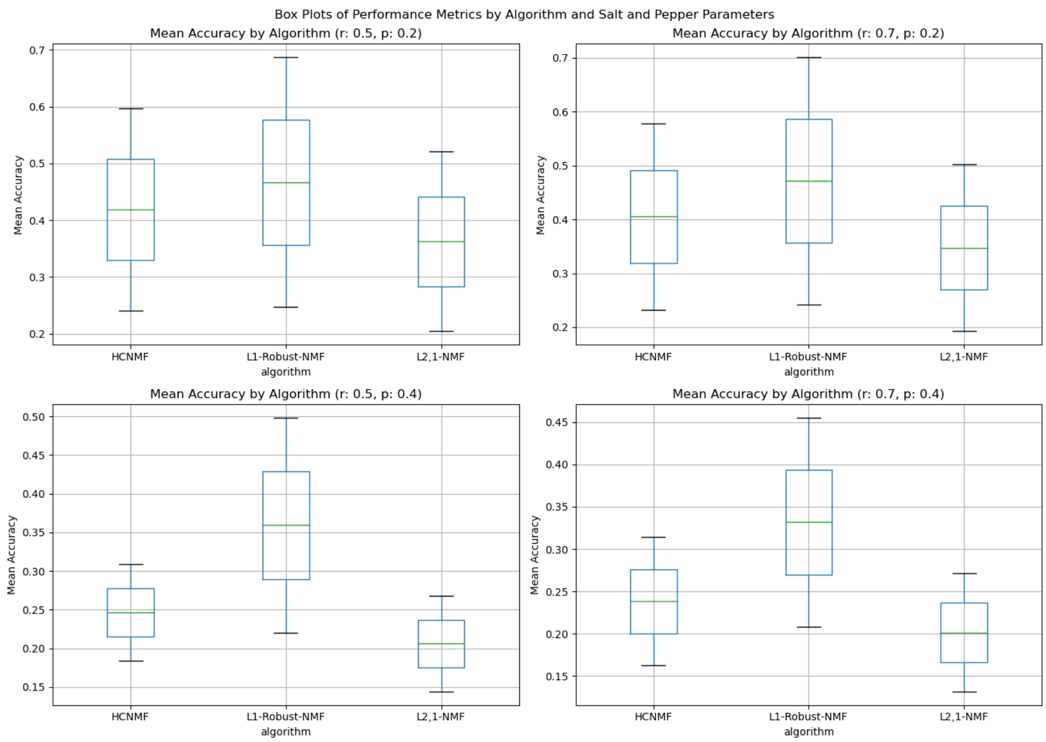

Figure 6: Mean Accuracy by algorithm, shown for different Salt and Pepper noise parameters (p and r).

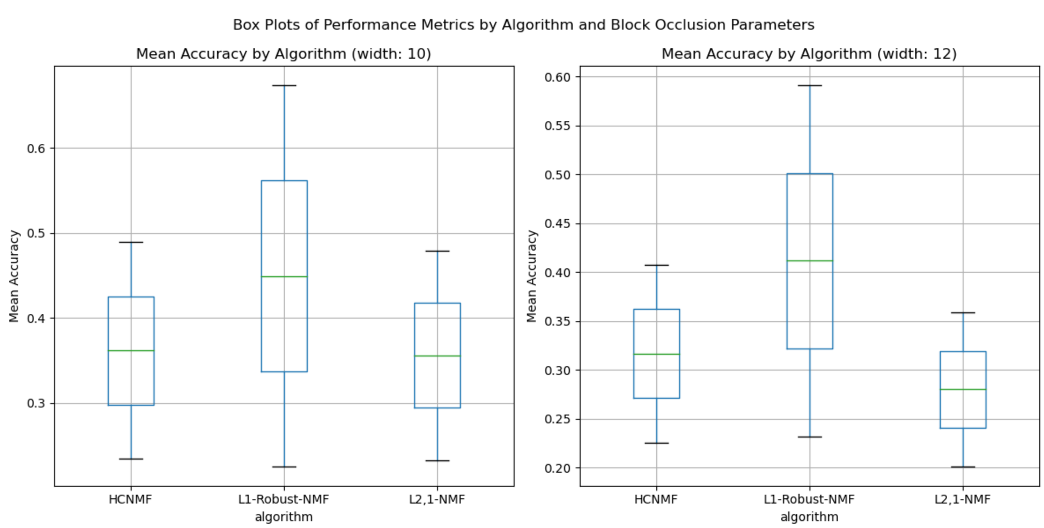

Figure 7: Mean Accuracy by algorithm, shown for different Block Occlusion noise sizes.

5 Conclusion

This study has compared three robust Non-negative Matrix Factorisation algorithms—L1-Regularised Robust NMF, L2,1-Norm Based NMF, and Hypersurface Cost Based NMF (HC-NMF)—for denoising and feature extraction on face image datasets corrupted by realistic noise patterns. Through experiments on the ORL and Extended YaleB datasets with salt-and-pepper and block occlusion noise at multiple intensity levels, we have demonstrated that each algorithm exhibits distinct strengths suited to different noise characteristics and dataset properties.

L1-Regularised Robust NMF achieved the lowest median reconstruction error and highest clustering accuracy and NMI scores. Its explicit modelling of sparse errors through L1 regularisation makes it effective for pixel-wise corruptions such as salt-and-pepper noise, reaching 70% clustering accuracy and 0.8 NMI on the ORL dataset. L2,1-Norm Based NMF demonstrated the most consistent performance, exhibiting the lowest standard deviation in clustering accuracy. Its sample-level error weighting proved effective for block occlusion noise, achieving slightly lower reconstruction error than L1-Robust-NMF under these conditions. HCNMF, despite higher reconstruction errors by Frobenius norm standards, achieved the highest accuracy (approximately 25%) and NMI (0.33) on the challenging Extended YaleB dataset with extreme illumination variations. Its bounded-influence cost function does not assume a specific error structure, making it suited for complex, non-sparse noise patterns.

The consistent relative performance rankings across varying noise parameters confirm that the observed strengths and weaknesses are fundamental to each algorithm's design principles. Our findings provide guidance for algorithm selection: L1-Robust-NMF for sparse, pixel-wise noise when reconstruction fidelity is critical; L2,1-NMF for structured occlusions and consistent performance; and HCNMF for complex scenarios with diverse noise characteristics.

Future work could explore hybrid approaches that combine these methods based on detected noise characteristics, investigate adaptive parameter selection strategies for λ_1, λ_2, and δ, and extend evaluation to other imaging modalities such as medical imaging where noise robustness is critical for diagnosis.

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A Code Structure and Execution Instructions

This project implements and compares three robust Non-negative Matrix Factorization (NMF) algorithms for image denoising and clustering on the ORL and ExtendedYaleB face datasets. The code is organized into five distinct Python files, and the experiments are configured to run in parallel for efficiency.

A.1 Project Structure

The project consists of five main Python files:

A.1.1 `tools.py`

This script contains all utility functions required for the experiments. This includes functions for loading data, adding salt-and-pepper and block occlusion noise, and evaluating clustering performance using accuracy (ACC) and Normalized Mutual Information (NMI).

A.1.2 `L21_nmf.py`

This file contains the implementation of the $L_{2,1}$-Norm Based NMF algorithm, which leverages $L_{2,1}$ norm regularization for robustness against sample-specific outliers. It includes the `L21NMF` class and an experiment runner function that saves results to JSON files.

A.1.3 `hcnmf.py`

This file implements the Hypersurface Cost-based NMF (HCNMF) algorithm. This method uses a robust hypersurface cost function to mitigate the influence of outliers. The file includes the core optimization function, reconstruction utilities, and the experiment runner.

A.1.4 `L1_Robust.py`

This script implements the L1-Regularized Robust NMF. The algorithm is designed to minimize the L1 reconstruction error, incorporating L1 regularization on the factor matrices and using a Huber loss function for smoothness. It contains the `L1RegularizedRobustNMF` class and its corresponding experiment function.

A.1.5 `main.py`

This is the main executable script that orchestrates the entire experimental suite. It iterates over all combinations of algorithms, datasets, and noise configurations, running them concurrently using Python's multiprocessing library to save time.

A.2 How to Run the Code

Please follow the steps below to set up and run the experiments.

1. **Install Dependencies:** Ensure the following Python libraries are installed: `numpy`, `scikit-learn`, `matplotlib`, and `pillow`.
2. **Prepare Datasets:** Place the two face datasets into the `data/` directory with the following structure:
    - The ORL dataset should be located at `data/ORL/`.

- The Extended YaleB dataset should be located at `data/CroppedYaleB/`.

3. **Execute Experiments:** Run the main script from your terminal:

   ```
   python main.py
   ```

   This command will start the experiments for all three NMF algorithms across both datasets and all configured noise types and parameters.

4. **Collect Results:** The script will automatically create a `results/` directory and save the output of each experimental run as a separate JSON file. You can monitor the progress of the parallel processes via the console output.

Note: The full suite of experiments is computationally intensive and may take up to 10 hours to complete, depending on your hardware. You can adjust parameters such as `n_runs` or `max_iter` within `main.py` to run a smaller-scale version of the experiments.